

AWS DataBrew Lab: Introduction to Data Cleaning

LAB OBJECTIVE

In this lab, you will:

- Identify and analyze data quality issues using a Data Profile
 - Fix inconsistent, missing, and invalid data in a visual DataBrew Project
 - Create a cleaned dataset ready for analytics
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STEP-BY-STEP LAB GUIDE

STEP 1: Access AWS Glue DataBrew

1. Sign in to the AWS Management Console
2. In the search bar, type "DataBrew" and open AWS Glue DataBrew
3. On the left sidebar, click **Datasets**

STEP 2: Create a New Dataset

1. Click "Connect new dataset"
2. **Dataset Name:** `employees`
3. Scroll to "Connect to new dataset → Amazon S3"

In the S3 path, enter:

`s3://databrew-data-lab-[UNIQUE-STRING]/raw-data/employee_data.csv`

4. Scroll down and click "Create dataset"

STEP 3: Create a Data Profile Job

Before cleaning, create a Data Profile Job to analyze your dataset and detect quality issues.

1. In the left navigation, click **Jobs**, then from the tab click **Profile jobs**
2. Click "Create job"
3. **Job Name:** `employee-data-profile`
4. Scroll to → **Job input**
5. Click "Browse datasets"
6. **Select dataset:** `employees`
7. Click on "Choose"
8. On **Job run sample** choose "Full dataset"

9. **Job output settings:**

s3://databrew-data-lab-[UNIQUE-STRING]/profile-results/ *(This will create a new folder to store the result)*

10. Scroll down to end to "Permissions"

11. Click "Create new IAM role", enter suffix [Name]-ID, and select it

12. Click "Create and run job"

13. On the "Job run history" tab

14. Wait until "Last job run status" changes to **Succeeded**

Once the job completes:

- Click on Job —> Profile Jobs
- Select the "employee-data-profile" Job name and click on "View data profile"
- Review:
 - % of missing values
 - Data types detected
 - Invalid values (e.g., date 2023-13-01)
 - Inconsistent capitalization
 - Unique vs duplicate records

Purpose: This helps you understand what issues to fix next.

STEP 4: Create a New Project (for Cleaning)

Now that you've profiled the data, it's time to clean it.

1. Click **Projects** in the left navigation
2. Click "Create project"
3. **Project Name:** employee-data-cleaning
4. Scroll down to **Select Dataset** choose employees
5. Under **Permissions**, choose the same role
(AWSGlueDataBrewServiceRole-[Name]-ID)
6. Click "Create project"

DataBrew will open your dataset in the **Recipe editor**, ready for transformation.

STEP 5: Explore Your Data

1. Review the raw dataset visually
2. Explore the built-in transformation
3. Identify clear problems:
 - Inconsistent capitalization
 - Missing values
 - Extra spaces
 - Mixed date and phone formats

BUILDING A RECIPE

STEP 6: Fix Capitalization in Names

1. Click the **name** column header
2. From the transformations menu
3. Choose "Format → Change to capital case"
4. On the right Format column pane scroll down
5. Click "Preview changes"
6. Click Apply ☒ All names become Proper Case (e.g., John Smith)

STEP 7: Standardize Department Names

1. Click **department** column header
2. From the transformations menu
3. Choose "Format → Change to lowercase"
4. On the right Format column pane scroll down
5. Click "Preview changes"
6. Click Apply ☒ All Department names become lowercase

STEP 8: Standardize email address

1. Click **email** column header
2. From the transformations menu
3. Choose "Format → Change to lowercase"
4. On the right Format column pane scroll down
5. Click "Preview changes"
6. Click Apply ☒ All email addresses become lowercase

STEP 8: Handle Missing Values

1. Click the **email** column header
2. From the transformations menu
3. Choose "Missing"
4. Then choose "Remove missing rows"
5. In the Missing value action choose "Delete rows with missing values"
6. Preview changes and Apply"
7. Repeat for **salary** column header
8. Repeat for **hire_date** column header

STEP 10: Fix Invalid or Inconsistent Dates

1. Click **hire_date** column header
2. From the transformations menu
3. Choose "Format → Date-time formats"
4. Select the format **YYYY-MM-DD**
5. On the right Format pane scroll down
6. Click "Preview changes"
7. Click Apply ☒ All valid dates now follow one consistent format

8. Also notice we have an invalid date `[2023-13-01]`
9. To fix this the click `hire_date` column header again
10. From the transformations menu
11. Choose "Invalid → Replace with custom value"
12. On the right Invalid pane scroll down
13. In Custom value textbox input the valid date `[2023-12-01]`
14. Scroll down Click "Preview changes"
15. Click Apply ☒ All dates are valid

STEP 11: Review Recipe Steps

1. On the right sidebar you can see the Recipe
2. Review transformations:

Each step is recorded and reusable.

STEP 12: Run the Transformation Job

1. Click "Create job" from the top right menu
2. **Job Name:** `employee-cleaned-job`
3. Scroll down to "Job Output Setting"
4. **Format Type:** CSV
5. **s3 Location:** `s3://databrew-data-lab-[UNIQUE-STRING]/cleaned-data/`
6. Scroll down to "Permissions" and choose the correct IAM role(
`AWSGlueDataBrewServiceRole-[Name]-ID`)
7. Click "Create and run job"
8. You can monitor the job in the JOBS section of the left navigation pane. The job will run, and a link to the output file will be placed in the Job output column.
9. Once the job changes its status to `Succeeded`, you can select the Job output link to show the location of the output file.
10. This will take you to the Amazon S3 service, where you can view the output file in the Amazon S3 console.

KEY CONCEPTS LEARNED

Data Quality Issues:

- Inconsistent capitalization
- Missing values
- Invalid entries and dates
- Mixed formats

Data Preparation Workflow:

1. Profile data to detect issues
 2. Standardize values and formats
 3. Handle missing or invalid records
 4. Export clean, reliable data
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